PYTHON FOUNDATIONS - ASSIGNMENT 01

User Input, Variables & Data Types

10 realistic mini-programs designed to make beginners combine multiple concepts in one problem.

input() Variables int() float() str() type() Arithmetic

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK

10 USE-CASE CHALLENGES

How to attempt this assignment

Objective: Build small interactive Python programs. Every question deliberately asks for several outputs so you learn how to combine input, variables, data types, type conversion and calculations.

1 Read first.
Identify every value the user must enter.

2 Choose types.
Decide which values should be `str`, `int` or `float`.

3 Convert input.
Remember that `input()` returns text by default.

4 Calculate.
Create new variables for totals, averages and derived values.

5 Display clearly.
Print meaningful labels with every answer.

6 Verify types.
Use `type()` whenever the question asks you to prove a data type.

Rules for this assignment: Do not use `if`, loops, lists, dictionaries or functions. Solve everything using only the concepts covered so far.

Before coding each question

- Write the required inputs.
- Predict the data type of each input.
- Mark where conversion is needed.
- Write the calculation formulas.
- Create meaningful variable names.
- Predict the final output format.

Teacher tip: Students should explain why they used `int()` or `float()` before running the code.

1 Student Profile & Marks Dashboard

INPUT | INT/FLOAT | AVERAGE | TYPE()

A school wants a tiny program that collects a student's basic details and creates a simple marks summary.

1. Take the student's name, age, class name, Maths marks, Science marks and English marks from the user.
2. Convert age to an integer. Choose a suitable numeric type for marks.
3. Calculate total marks out of 300.
4. Calculate the average marks.
5. Calculate the student's age after 5 years.
6. Print a clean student summary containing all original and calculated values.
7. Use `type()` to display the data types of name, age and average.

Stretch: Also ask for monthly pocket money and calculate the amount received in one year.

2 Online Shopping Bill

PRICE | QUANTITY | MULTIPLICATION | CONVERSION

An online store needs a basic checkout calculator for one product.

1. Ask for customer name, product name, product price and quantity.
2. Store product price as `float` and quantity as `int`.
3. Calculate the subtotal: price x quantity.
4. Ask the user to enter a delivery charge and add it to the subtotal.
5. Ask for an advance amount already paid and subtract it from the bill.
6. Print subtotal, delivery charge, final bill and remaining amount to pay.
7. Display the data types of product name, price, quantity and final bill.

Stretch: Ask for the price of a gift-wrap service and include it in the final bill.

3 Food Delivery Order Summary

REAL-LIFE BILL | FLOAT | TOTAL

A food delivery app wants to show a simple order summary before checkout.

1. Take customer name, restaurant name, item name, price per item and quantity.
2. Ask separately for packaging charge and delivery charge.
3. Convert price and charges to `float`, and quantity to `int`.
4. Calculate food cost: price per item x quantity.
5. Calculate the complete payable amount by adding packaging and delivery.
6. Ask how many people are sharing the order and calculate cost per person.
7. Print an order summary and the data types of quantity and payable amount.

Think: Which operation should you use to divide the bill equally among people?

4 Road Trip Fuel Planner

DISTANCE | MILEAGE | DIVISION | FLOAT

A family wants a quick estimate of how much fuel and money a road trip may require.

1. Take traveller name, destination, one-way distance in km, vehicle mileage in km/litre and fuel price per litre.
2. Convert distance, mileage and fuel price to `float`.
3. Calculate round-trip distance.
4. Calculate litres of fuel required: round-trip distance / mileage.
5. Calculate total estimated fuel cost.
6. Ask the number of travellers and calculate fuel cost per traveller.
7. Print all results with meaningful labels and verify the type of total fuel cost.

Stretch: Ask for toll charges and add them to the trip cost before calculating per-person cost.

5 Employee Salary Snapshot

MONTHLY/YEARLY | FLOAT | SUBTRACTION

An employee wants a very simple personal salary calculator. This is only arithmetic - no tax rules are required.

1. Take employee name, company name and monthly salary.
2. Ask for monthly transport allowance and monthly food allowance.
3. Calculate total monthly income including both allowances.
4. Calculate total yearly income by multiplying monthly income by 12.
5. Ask for monthly personal expenses and calculate yearly personal expenses.
6. Calculate yearly amount left after personal expenses.
7. Print the salary snapshot and show types of monthly salary and yearly amount left.

Think: Money can contain decimal values. Which numeric type is more flexible here?

6 Movie Night Ticket Calculator

TICKETS | SNACKS | TOTAL | PER PERSON

A group of friends is planning a movie night and wants to know the total and per-person cost.

1. Ask for movie name, theatre name, ticket price and number of tickets.
2. Ask for total snack cost and parking cost.
3. Convert number of tickets to `int`; convert prices/costs to `float`.
4. Calculate total ticket cost.
5. Calculate the complete movie-night cost including snacks and parking.
6. Calculate average cost per person using the number of tickets.
7. Print a complete summary and use `type()` on movie name, ticket count and average cost.

Stretch: Ask for an amount already contributed by one friend and print the remaining group amount.

7 Mobile Data Usage Calculator

DATA PLAN | MULTIPLICATION | SUBTRACTION

A mobile user wants a simple data-usage summary for the current plan period.

1. Take user name, network provider, daily data allowance in GB and number of plan days.
2. Calculate total data available for the complete plan.
3. Ask how much data in GB has already been used.
4. Calculate remaining data.
5. Ask for the plan price and calculate average cost per day.
6. Calculate average cost per GB using total plan data.
7. Print a clean plan summary and display types of plan days, used data and provider name.

Stretch: Ask for extra data purchased in GB and add it to the total available data.

8 School Trip Budget

BUDGET | MULTIPLE COSTS | SHARING

A teacher is planning a class trip and needs a basic budget estimate.

1. Take school name, destination and number of students.
2. Ask for bus cost, entry-ticket cost per student and meal cost per student.
3. Calculate total entry-ticket cost.
4. Calculate total meal cost.
5. Calculate total trip cost including the bus.
6. Calculate the equal cost per student if the whole trip cost is shared equally.
7. Print the complete budget and show the types of student count and cost per student.

Think: Student count should normally be a whole number, while costs may contain decimals.

9 Daily Water & Activity Tracker

HEALTHY HABITS | UNIT CALCULATION | AVERAGE

Create a simple arithmetic-only daily activity summary. Do not make health recommendations or decisions.

1. Take user name, number of glasses of water, water per glass in ml and walking minutes.
2. Calculate total water consumed in ml.
3. Convert the total water amount to litres by dividing by 1000.
4. Ask the number of walking sessions completed and calculate average minutes per session.
5. Ask yesterday's walking minutes and calculate the numerical difference between today and yesterday.
6. Print the full activity summary.
7. Use `type()` to display the types of name, glasses and litres consumed.

Stretch: Ask for sleep hours for two nights and calculate the simple average.

10 Mini Bank Account Summary

BALANCE | DEPOSIT | WITHDRAWAL | TYPE()

Build a basic arithmetic account summary. There is no login, validation or banking rule in this exercise.

1. Take account-holder name and an account nickname such as "Savings".
2. Take opening balance, deposit amount and withdrawal amount from the user.
3. Convert all money values to `float`.
4. Calculate balance after deposit.
5. Calculate final balance after withdrawal.
6. Ask for a monthly savings contribution and calculate the amount contributed over 12 months.
7. Print the complete account summary and show the data types of holder name, opening balance and final balance.

Stretch: Create one more variable that adds the 12-month contribution amount to the current final balance and print it as a simple projected total.

Before you submit

For every question, your program should be understandable even to someone seeing it for the first time. Good programmers do not only make code run - they make the code readable.

- I used meaningful variable names.
- I used `input()` for values that the user must enter.
- I converted numeric input using `int()` or `float()` where required.
- I did not accidentally put quotes around variables while printing their values.
- I created separate variables for important calculations.
- I printed labels such as "Total Bill:" instead of printing unexplained numbers.
- I used `type()` wherever the question requested data-type verification.
- I tested my program with at least two different sets of inputs.
- I did not use `if`, loops, lists, dictionaries or functions.

Bonus classroom activity: After completing any two questions, exchange programs with a classmate. Without changing the code, explain what data type each variable contains and how each calculated value is produced.

Assignment Complete

Questions attempted: _____ / 10 Teacher review: _____